Numerical Analysis chapter 4 theoretical homework

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Due time: November 25, 2024

I

$1.11011101 * 2^8$

II

$1.00110011 \cdots * 2^{-1}$

III

Proof: We donate $x = 1.0 \cdots 0 * \beta^e$, then we have

$$x - x_L = 1 * \beta^{e-1} * \beta^{-p}, x_R - x = 1 * \beta^e * \beta^{-p}.$$

So, we have $\beta(x - x_L) = x_R - x$.

IV

We have 0 01111110 00110011001100110011010 $= x_R > \frac{2}{3} > x_L =$ 0 01111110 00110011001100110011001, the relative error is $3.97 * 10^{-8}$.

V

The older version of $\epsilon_u = \frac{1}{2} * 2^{-23} = 2^{-24}$, and the new version of $\epsilon_u = 2^{-23}$.

VI

We need to calculate that $1 - cos(\frac{1}{4})$. We know that $cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \cdots$, so we have $1 - cos(\frac{1}{4}) = \frac{1}{32} - \frac{1}{6144} + \cdots$. During the calculation, we have lost the precision of the last term. We lost $log_2(\frac{1}{32} - \frac{1}{6144}) = 5$ digits of precision.

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VII

- $1 - cos(x) = \frac{x^2}{2} - \frac{x^4}{4!} + \cdots$
- $1 - cos(x) = 2sin^2(\frac{x}{2})$

VIII

As we all know, $\kappa(x) = |\frac{xf'(x)}{f(x)}|$.

a

$f(x) = (x-1)^{\alpha}, f'(x) = \alpha(x-1)^{\alpha-1}$. $\kappa(x) = |\frac{\alpha x}{x-1}|$, so when $x \to 1$, $\kappa(x) \to \infty$.

b

$f(x) = ln(x), f'(x) = \frac{1}{x}$, we have $\kappa(x) = |\frac{1}{lnx}|$, so when $x \to 1$, $\kappa(x) \to \infty$.

c

We have $\kappa(x) = |x|$, so when $x \to \infty$, $\kappa(x) \to \infty$.

d

$f(x) = arccos(x), f'(x) = -\frac{1}{\sqrt{1-x^2}}$, we have $\kappa(x) = |\frac{x}{arccos(x)*\sqrt{1-x^2}}|$, so when $x \to \pm 1$, $\kappa(x) \to \infty$.

IX

$f(x) = 1 - e^{-x}, x \in [0, 1]$

1

We donate that $a(x) = cond_f(x) = |\frac{xf'(x)}{f(x)}| = |\frac{xe^{-x}}{1-e^{-x}}| = \frac{x}{e^x-1}.a'(x) = \frac{e^x-1-xe^x}{(e^x-1)^2}$, which means that $a(x)$ is increasing in $[0, 1]$. So, $a(0) = 0, a(1) = \frac{1}{e-1} < 1$, which indicates that $cond_f(x)$ is bounded when $x \in [0, 1]$.

2

From **Theorem 4.78** and question 1, we could know that $f_A(x) = f(x)(1 + \delta(x)), |\delta(x)| \le \varphi(x)\varepsilon_u$, and $cond_A(x) \le \frac{\varphi(x)}{cond_f(x)}$. $f_A(x) = fl[fl(1 - fl(e^{-fl(x)}))], |\delta_i < \varepsilon_u|$. We know that $e^{-x\delta_1} = 1 - x\delta_1$.

$$\begin{aligned} f_A(x) &= (1 - e^{-x(1+\delta_1)}(1+\delta_2))(1+\delta_3) \\ &= (1 - e^{-x})(1 + \delta_2 + \delta_3 - x\delta_1 - \frac{\delta_2 - x\delta_1}{1 - e^{-x}}) \end{aligned}$$

We donate $\varphi(x) = 3 + \frac{2}{1-e^{-x}}$. $\therefore cond_A(x) \le \frac{\varphi(x)}{cond_f(x)} = \frac{5-3e^{-x}}{xe^{-x}}$

3

Below are figures of $cond_f$ and $cond_A$

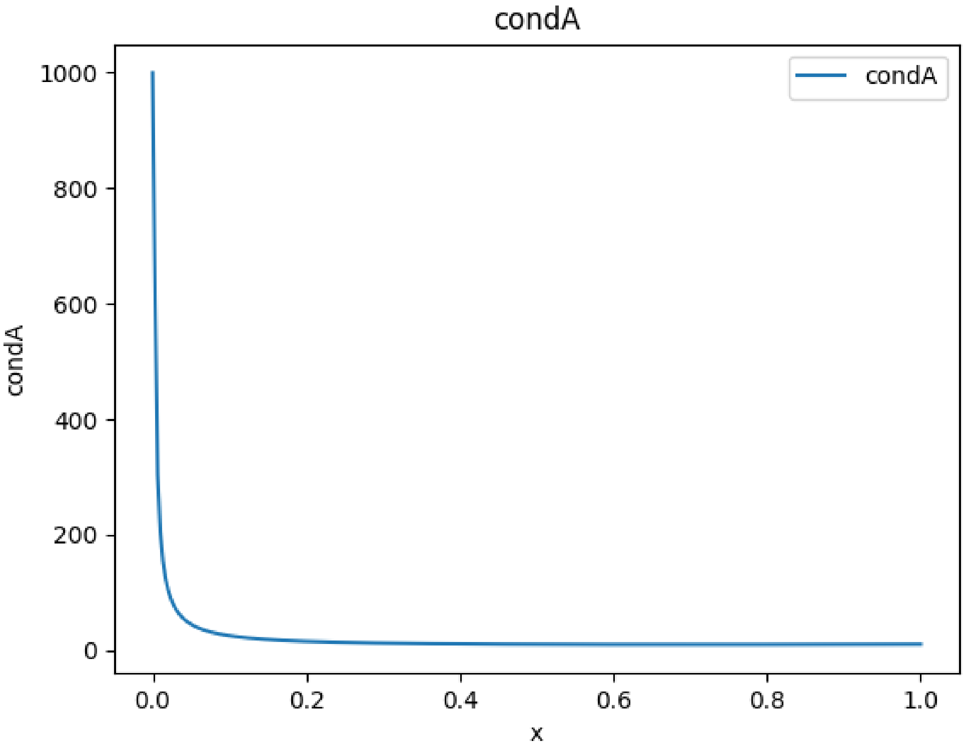

Figure 1: The graph of $cond_A$

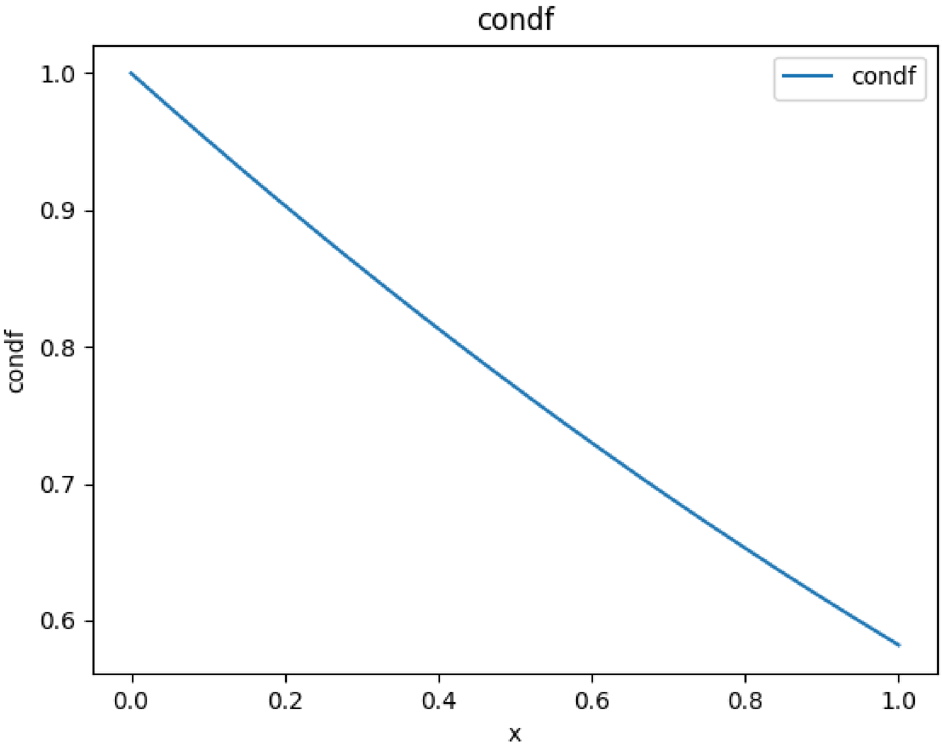

Figure 2: The graph of $cond_f$

The result is generated by $Python3$.

X

We know that $\|A\|_2 = \sigma_{max}(A)$, also it is easy to know that $\|A^{-1}\|_2 = \sigma_{max}(A^{-1}) = \frac{1}{\sigma_{min}(A)}$. For the unity matrix, we have $\|I\|_2 = 1$, and $\|I^{-1}\|_2 = 1$, which means that $cond_2(A) = 1$.

XI

$q(x) = \sum_{i=0}^{n} a_i x_i, a_n = 1$. $cond_f(x) = |\frac{1}{r}| \cdot \sum_{i=0}^{n-1} |a_i \frac{\partial r}{\partial a_i}|$,and $\kappa_i(f) = |\frac{1}{r}| \cdot |a_i \frac{\partial r}{\partial a_i}|$. Now we calculate $\frac{\partial r}{\partial a_i}$, we have $\frac{\partial q}{\partial a_i} + \frac{\partial q}{\partial r} \cdot \frac{\partial r}{\partial a_i} = 0$. So, we know that $\frac{\partial r}{\partial a_i} = -\frac{r^i}{\sum_{j=1}^{n} j a_j r^{j-1}}$ $\kappa(x) = max|\frac{r^{i-1} a_i}{p'(r)}|$. Now follow the Wilkinson example,$q(r) = \prod_{i=1}^{n}(x-i)$. $\frac{\mathrm{d}}{\mathrm{d}x} \prod_{i=1}^{n}(x-i) = \sum_{j=1}^{n} \prod_{i \neq j}^{n}(x-i)$. $\kappa_i(x) = |\frac{r^i a_i}{r \sum_{j=1}^{n} \prod_{i \neq j}^{n}(x-i)}|$

$$cond_f(x) = max_i|\frac{a_i p^{i-1}}{(p-1)!}| \geq \frac{\sum_{k=1}^{p} kp^{p-2}}{2(p-1)!} = \frac{(p+1)p^{p-1}}{2(p-1)!}$$

XII

We have $(2, 2, -1, 1)$, $a = (1.0)_2 * 2^0, b = (1.1)_2 * 2^0$. We have $\frac{a}{b} = \frac{2}{3} = (0.1010\cdots)$. When we choose the precision of $2p = 4$: $(0.101)_2 = 1.01 * 2^{-1}$. $fl(\frac{a}{b}) = (1.0)_2 * 2^{-1} = \frac{1}{2}$. Now, we calculate the relative error: $E_{rel} = \frac{1}{4} = 2^{-2} = \frac{1}{2} * 2^{1-2} = \varepsilon_u$, which contradicts the conclusion of the model of machine arithmetic.

XIII

According to **IEEE754**, we have learnt that there are 24 bits for the mantissa. Thinking about the root in range of $[128, 129]$, it will be represented as $1.xxx \cdots * 2^7$, the width of the root is $2^{-n} < 10^{-6}$,i.e. $n = 20$.

- The spacing between consecutive single-precision FPNs in this range is determined by the exponent:

$$\text{Spacing} = 2^{7-23} = 2^{-16} \approx 1.5 \times 10^{-5}.$$

- This means the smallest representable interval width in this range is approximately 1.5×10^{-5}, which is much larger than 10^{-6}.

So, we cannot compute the root with absolute accuracy $< 10^{-6}$.

XIV

For $s(x) = ax^3 + bx^2 + cx + d$, we need to know the values of $s(x), s'(x)$ at the point x_i, x_{i+1}. Note that $s'(x) = 3ax^2 + 2bx + c, a, b, c, d$ can be calculated by

$$\begin{cases} ax_i^3 + bx_i^2 + cx_i + d = f(x_i) \\ ax_{i+1}^3 + bz_{i+1}^2 + cx_{i+1} + d = f(x_{i+1}) \\ 3ax_i^2 + 2bx_i + c = f'(x_i) \\ 3ax_{i+1}^2 + 2bx_{i+1} + c = f'(x_{+1}) \end{cases} \Leftrightarrow \begin{bmatrix} x_i^3 & x_i^2 & x_i & 1 \\ x_{i+1}^3 & x_{i+1}^2 & x_{i+1} & 1 \\ 3x_i^2 & 2x_i & 1 & 0 \\ 3x_{i+1}^2 & 2x_{i+1} & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} f(x_i) \\ f(x_{i+1}) \\ f'(x_i) \\ f'(x_{i+1}) \end{bmatrix}$$

Denote the cofficients matrix as A, then the maximal condition number is

$$\max cond_f(x) = \|A\|_2 \left\|A^{-1}\right\|_2 = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

When $x_i \to x_{i+1}$, we have find that $\lambda_{min} \to 0$ so the condition number is ∞, whcih means in that case that the problem is ill-conditioned.

Acknowledgement